
EP 403: Physics of atoms and molecules

Course Code	EP 403
Course Name	Physics of atoms and molecules
L-T-P-C	3-0-0-3
Prerequisites	PH301 and/or faculty consent; Out of EP403 and PH524, a student is allowed to take only one course.
Intended for	UG
Distribution	Core
Approval	30th Senate (Feb 2021), in the B.Tech Engineering Physics curriculum; recommended by the 40th BoA (May 2021)

Course Contents

- **Module I**

Time-independent perturbation theory for first and second-order correction and its application; Identical particles, eigen functions for identical two particle systems. [6 Lectures]

- **Module II**

Time-dependent perturbation theory for two level systems and its generalization to higher level systems, its application to Fermi-Golden rule. [6 Lectures]

- **Module III**

Interaction of electromagnetic radiation with single electron atoms, Sinusoidal perturbation and Rabi flopping, Dipole approximation and dipole selection rules, Transition rates, Line broadening mechanisms, Incoherent perturbations, spontaneous and stimulated emissions and Einstein coefficients, selection rules for transitions [12 Lectures]

- **Module IV**

Review of atomic structure of H, Atomic structure of two electron system-variational method, central field approximation, Slater determinant, L-S coupling, J-J coupling. [7 Lectures]

- **Module V**

General nature of molecular structure, molecular binding, LCAO, Born-Oppenheimer Approximation. [5 Lectures]

- **Module VI**

Introduction to infra-red and Raman spectroscopy, Introduction to group theory and symmetry and Spectroscopy. [6 Lectures]

Text Books

1. C. J. Foot, **Atomic Physics**, Oxford, First edition 2005.
2. C. Banwell and E. Maccash, **Fundamentals of molecular spectroscopy**, Mc Graw Hill, 2013.